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United States Patent	12398913
Kind Code	B2
Date of Patent	August 26, 2025
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Adjustable diffuser assembly systems and methods

Abstract

A diffuser assembly for a heating, ventilation, and air conditioning (HVAC) system includes a housing having a wall and a passage formed in the wall. The diffuser assembly also includes a first adjustment panel moveably coupled to the wall and having a first flange and a second adjustment panel moveably coupled to the wall and having a second flange. The first flange and the second flange define at least a portion of an inlet port of the diffuser assembly that extends through the passage. The inlet port is configured to fluidly couple to a duct and direct an air flow from the duct into an interior volume of the housing. The first adjustment panel and the second adjustment panel are configured to translate along the wall to adjust a size of the inlet port.

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Appl. No.: 17/562836

Filed: December 27, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220205677 A1	Jun. 30, 2022

Related U.S. Application Data

us-provisional-application US 63131222 20201228

Publication Classification

Int. Cl.: F24F13/06 (20060101); F24F13/02 (20060101); F24F13/10 (20060101); F24F13/32 (20060101)

U.S. Cl.:

CPC F24F13/06 (20130101); F24F13/0209 (20130101); F24F13/10 (20130101); F24F13/32 (20130101); F24F2221/00 (20130101)

Field of Classification Search

CPC: F24F (13/06); F24F (13/0209); F16L (25/14); F16L (41/12); F16L (21/06); F16L (5/00)

USPC: 285/67; 285/419

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority from and the benefit of U.S. Provisional Application No. 63/131,222, entitled “ADJUSTABLE AIR INLET DIFFUSER,” filed Dec. 28, 2020, which is herein incorporated by reference in its entirety for all purposes.

BACKGROUND

(1) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

(2) Heating, ventilation, and air conditioning (HVAC) systems are utilized in residential,

commercial, and industrial environments to control environmental properties, such as temperature and humidity, for occupants of the respective environments. The HVAC system may regulate the environmental properties through delivery of a conditioned air flow to the environment. For example, the HVAC system generally includes an HVAC unit that is fluidly coupled to various rooms or spaces within the building via an air distribution system, such as a system of ductwork. The HVAC unit may be operable to direct a heated air flow or a cooled air flow through the ductwork and into the spaces to be conditioned. In this manner, the HVAC unit facilitates regulation of environmental parameters within the rooms or spaces of the building. Generally, one or more diffuser assemblies are fluidly coupled to the ductwork and are configured to facilitate distribution of the conditioned air into the rooms or spaces to be conditioned. Unfortunately, it may be arduous, costly, and/or time consuming to manufacture, adjust, and/or install different diffuser assemblies configured to couple to various sizes of ductwork that may be installed in the building serviced by the HVAC system.

SUMMARY

(3) A summary of certain embodiments disclosed herein is set forth below. It should be noted that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(4) The present disclosure relates to a diffuser assembly for a heating, ventilation, and air conditioning (HVAC) system. The diffuser assembly includes a housing having a wall and a passage formed in the wall. The diffuser assembly also includes a first adjustment panel moveably coupled to the wall and having a first flange and a second adjustment panel moveably coupled to the wall and having a second flange. The first flange and the second flange define at least a portion of an inlet port of the diffuser assembly that extends through the passage. The inlet port is configured to fluidly couple to a duct and direct an air flow from the duct into an interior volume of the housing. The first adjustment panel and the second adjustment panel are configured to translate along the wall to adjust a size of the inlet port.

(5) The present disclosure also relates to a diffuser assembly for a heating, ventilation, and air conditioning (HVAC) system. The diffuser assembly includes a wall having a passage extending therethrough. The diffuser assembly also includes a first adjustment panel moveably coupled to the wall and having a first flange extending into the passage. The diffuser assembly includes a second adjustment panel moveably coupled to the wall and having a second flange extending into the passage. The first flange and the second flange define at least a portion of an inlet port extending through the passage and configured to fluidly couple to a duct. The first adjustment panel and the second adjustment panel are configured to translate along the wall between a first configuration to define a first size of the inlet port and a second configuration to define a second size of the inlet port.

(6) The present disclosure also relates to a diffuser assembly for a heating, ventilation, and air conditioning (HVAC) system. The diffuser assembly includes a housing having a wall and a passage formed in the wall. The diffuser assembly includes a first adjustment panel having a first base portion moveably coupled to the wall. The first adjustment panel also includes a first flange extending from the first base portion into the passage. The diffuser assembly includes a second adjustment panel having a second base portion moveably coupled to the wall. The second adjustment panel also includes a second flange extending from the second base portion into the passage. The first flange and the second flange define at least a portion of an inlet port extending through the passage and configured to receive an air flow from a duct. The first base portion and the second base portion are configured to translate along the wall to expose or occlude a portion of the passage to adjust a size of the inlet port in the passage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:
- (2) FIG. 1 is a perspective view of an embodiment of a building incorporating a heating, ventilation, and/or air conditioning (HVAC) system in a commercial setting, in accordance with an aspect of the present disclosure;
- (3) FIG. 2 is a schematic of an embodiment of a room of a building having diffuser assemblies, in accordance with an aspect of the present disclosure;
- (4) FIG. 3 is a perspective view of an embodiment of a diffuser assembly, in accordance with an aspect of the present disclosure;
- (5) FIG. 4 is an exploded perspective view of an embodiment of a diffuser assembly, in accordance with an aspect of the present disclosure;
- (6) FIG. 5 is a cross-sectional perspective view of an embodiment of a portion of a diffuser assembly, in accordance with an aspect of the present disclosure;
- (7) FIG. 6 is a cross-sectional perspective view of an embodiment of a portion of a diffuser assembly, in accordance with an aspect of the present disclosure;
- (8) FIG. 7 is an elevation view of an embodiment of a portion of a diffuser assembly having adjustment panels in an open configuration, in accordance with an aspect of the present disclosure;
- (9) FIG. 8 is perspective view of an embodiment of a diffuser assembly having adjustment panels in an expanded configuration, in accordance with an aspect of the present disclosure;
- (10) FIG. 9 is an elevation view of an embodiment of a portion of a diffuser assembly having adjustment panels in an intermediate configuration, in accordance with an aspect of the present disclosure;
- (11) FIG. 10 is an elevation view of an embodiment of a portion of a diffuser assembly having adjustment panels in a contracted configuration, in accordance with an aspect of the present disclosure;
- (12) FIG. 11 is perspective view of an embodiment of a diffuser assembly having adjustment panels in a contracted configuration, in accordance with an aspect of the present disclosure;
- (13) FIG. 12 is a perspective view of an embodiment of an adjustment panel of a diffuser assembly, in accordance with an aspect of the present disclosure;
- (14) FIG. 13 is a perspective view of an embodiment of a diffuser assembly, in accordance with an aspect of the present disclosure; and
- (15) FIG. 14 is a perspective view of an embodiment of adjustment panels of a diffuser assembly, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

- (16) One or more specific embodiments of the present disclosure will be described below. These described embodiments are only examples of the presently disclosed techniques. Additionally, in an effort to provide a concise description of these embodiments, all features of an actual implementation may not be described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.
- (17) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms

“comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(18) As used herein, the terms “approximately,” “generally,” “substantially,” and so forth, are intended to convey that the property value being described may be within a relatively small range of the property value, as those of ordinary skill would understand. For example, when a property value is described as being “approximately” equal to (or, for example, “substantially similar” to) a given value, this is intended to convey that the property value may be within $\pm 5\%$, within $\pm 4\%$, within $\pm 3\%$, within $\pm 2\%$, within $\pm 1\%$, or even closer, of the given value. Similarly, when a given feature is described as being “substantially parallel” to another feature, “generally perpendicular” to another feature, and so forth, this is intended to convey that the given feature is within $\pm 5\%$, within $\pm 4\%$, within $\pm 3\%$, within $\pm 2\%$, within $\pm 1\%$, or even closer, to having the described nature, such as being parallel to another feature, being perpendicular to another feature, and so forth. Mathematical terms, such as “parallel” and “perpendicular,” should not be rigidly interpreted in a strict mathematical sense, but should instead be interpreted as one of ordinary skill in the art would interpret such terms. For example, one of ordinary skill in the art would understand that two lines that are substantially parallel to each other are parallel to a substantial degree, but may have minor deviation from exactly parallel.

(19) As briefly discussed above, a heating, ventilation, and/or air conditioning (HVAC) system may be used to thermally regulate a space within a building, home, or other suitable structure. The HVAC system may include an HVAC unit configured to condition an air flow via an evaporator, a furnace, a heating coil, a chiller system, another heat exchange system, or a combination thereof, and to provide the conditioned air flow (e.g., a heated air flow, a cooled air flow, a dehumidified air flow) to the space. For example, the HVAC unit may be fluidly coupled to the space via an air distribution system, such as a system of ductwork, which extends between the HVAC unit and the space. One or more fans or blowers of the HVAC system may be operable to direct a supply of conditioned air from the HVAC unit, through the ductwork, and into the spaces within the building.

(20) Typically, the HVAC system includes one or more diffusers that are fluidly coupled to the ductwork and are configured to facilitate distribution of air from the ductwork into the rooms or spaces of the building. For example, the diffusers may be positioned adjacent to ceilings, floors, and/or walls of the rooms conditioned by the HVAC system and may be configured to discharge air from the ductwork into the rooms or other spaces. In many cases, the building serviced by the HVAC system may include ductwork with a variety of ducts having various sizes (e.g., cross-sectional areas and/or cross-sectional geometries), for example. Accordingly, multiple different diffuser assemblies may be manufactured that include dedicated or particularly-designed inlets configured to receive and couple to corresponding types (e.g., sizes) of ducts. That is, various different types of diffuser assemblies may be manufactured to accommodate and enable installation of the diffuser assemblies onto ducts having various sizes or other characteristics. Unfortunately, manufacturing multiple different diffuser assemblies that are each configured for installation on a particular size and/or shape of duct may increase costs of manufacture of the diffuser assemblies, increase storage costs associated with maintaining a relatively large quantity or inventory of different diffuser assemblies (e.g., prior to installation of the diffuser assemblies on a jobsite), and/or complicate selection and installation of the diffuser assemblies at a jobsite.

(21) It is presently recognized that enabling an individual diffuser assembly to couple (e.g., fluidly couple, mechanically couple) to multiple different types (e.g., sizes) of ducts may facilitate improved installation of the diffuser assembly at a jobsite, as well as reduce cost and/or complexity that may be associated with manufacturing and offering the diffuser assembly. Accordingly, embodiments of the present disclosure are directed toward an adjustable diffuser assembly that is

configured to couple to multiple different types (e.g., sizes) of ducts. For example, the adjustable diffuser assembly may include a housing having a passage that extends through a wall of the housing. The adjustable diffuser assembly may include a set (e.g., one or more) of adjustment panels that are moveably (e.g., slideably) coupled to the wall of the housing and include a set of flanges that extend into the passage. The flanges may define at least a portion of an inlet port of the adjustable diffuser assembly. The inlet port may facilitate coupling (e.g., fluidic coupling, mechanical coupling) of the diffuser assembly to a duct or duct outlet. The adjustment panels may, via translation along and relative to the housing of the diffuser assembly, enable adjustment of a size (e.g., a cross-sectional area, a cross-sectional shape) of the inlet port of the diffuser assembly. In particular, the adjustment panels may be adjustable to cause the size of the inlet port to substantially match or correspond with sizes or other characteristics of various different ducts. For example, an operator (e.g., a service technician installing the adjustable diffuser assembly at a jobsite) may transition the adjustment panels to a first configuration on or relative to the housing of the diffuser assembly in which the size of the inlet port formed by the flanges of the adjustment panels is relatively large. In this way, the relatively large inlet port may facilitate coupling of the adjustable diffuser assembly to a duct having a relatively large size. Alternatively, the operator may transition the adjustment panels to a second configuration on or relative to the housing of the diffuser assembly in which the size of the inlet port formed by the flanges of the adjustment panels is relatively small. In this way, the relatively small inlet port may facilitate coupling of the adjustable diffuser assembly to a duct having a relatively small size. As discussed below, it should be understood that the adjustment panels may be transitionable to a plurality of intermediate configurations between the first and second configurations to facilitate coupling of the inlet port to multitudinous different sizes and/or shapes of ducts. In this manner, the adjustable diffuser assembly disclosed herein may be configured to couple to a variety of different types of ducts. These and other features will be described below with reference to the drawings.

(22) Turning now to the drawings, FIG. 1 illustrates an embodiment of a heating, ventilation, and/or air conditioning (HVAC) system for environmental management that employs one or more HVAC units in accordance with the present disclosure. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an “HVAC system” as used herein is defined as conventionally understood and as further described herein. Components or parts of an “HVAC system” may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An “HVAC system” is a system configured to provide such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

(23) In the illustrated embodiment, a building **10** is air conditioned by an HVAC system **11** having an HVAC unit **12**. The building **10** may be a commercial structure or a residential structure. As shown, the HVAC unit **12** is disposed on the roof of the building **10**; however, the HVAC unit **12** may be located in other equipment rooms or areas adjacent the building **10**. The HVAC unit **12** may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit **12** may be part of a split HVAC system, which includes an outdoor HVAC unit and an indoor HVAC unit.

(24) The HVAC unit **12** is an air cooled device that implements a refrigeration cycle to provide conditioned air to the building **10**. Specifically, the HVAC unit **12** may include one or more heat

exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building **10**. In the illustrated embodiment, the HVAC unit **12** is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building **10**. The HVAC unit **12** may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with gas heat, or cooling with a heat pump. For example, in certain embodiments, the HVAC unit **12** may be a heat pump that provides both heating and cooling to the building with one refrigeration circuit configured to operate in different modes. In other embodiments, the HVAC unit **12** may include one or more refrigeration circuits for cooling an air stream and a furnace for heating the air stream.

(25) In any case, after the HVAC unit **12** conditions the air, the air may be supplied to the building **10** via ductwork **14** (e.g., an air distribution system) extending from the HVAC unit **12** and throughout the building **10**. For example, the ductwork **14** may extend to various individual floors, rooms zones, or other sections or spaces of the building **10**. In the illustrated embodiment, a plurality of diffuser assemblies **16** is coupled to the ductwork **14**. The diffuser assemblies **16** may direct the conditioned air into the various spaces of the building **10** in a manner that improves air distribution and/or air dispersion across the spaces.

(26) In some embodiments, a control device **18**, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air supplied by the HVAC unit **12**. The control device **18** also may be used to control the flow of air through the ductwork **14**. For example, the control device **18** may be used to regulate operation of one or more components of the HVAC unit **12** or other components, such as dampers and fans, within the building **10** that may control flow of air through and/or from the ductwork **14**. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of supply air, return air, and so forth. Moreover, the control device **18** may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building **10**.

(27) FIG. 2 is a schematic of an embodiment of a room **30** of the building **10**, illustrating diffuser assemblies **16** coupled to ductwork **14**. The diffuser assemblies **16** may be fluidly coupled to the ductwork **14** to receive a flow of conditioned air **32**, which may be generated by the HVAC unit **12**, for example. The diffuser assemblies **16** are also fluidly coupled to the room **30**. Accordingly, the diffuser assemblies **16** may discharge the conditioned air **32** into a space **34** of the room **30**. In some embodiments, the building **10** may include a dropped ceiling **36** (e.g., ceiling tiles) that may be suspended from a ceiling structure **38** of the building **10**. At least a portion of the ductwork **14** and the diffuser assemblies **16** may be located in a space formed between the ceiling structure **38** and the dropped ceiling **36**. In other embodiments, the ductwork **14** and/or the diffuser assemblies **16** may be located in any other suitable region of the building **10**. For example, the ductwork **14**, the diffuser assemblies **16**, or both, may be partially or fully integrated into the ceiling structure **38** of the building **10** and/or located in walls **40** or a floor **42** of the building **10**.

(28) FIG. 3 is a perspective view of an embodiment of one of the diffuser assemblies **16**, referred to herein as a diffuser assembly **50** (e.g., an adjustable diffuser assembly). In the illustrated embodiment, the diffuser assembly **50** includes a housing **52** having a first wall **54** (e.g., a front wall), a second wall **56** (e.g., a rear wall), and a web **58** (e.g., lateral wall, lateral sides, wrap, etc.) that extends between (e.g., from and to) the first wall **54** and the second wall **56**. The first wall **54**, the second wall **56**, and the web **58** may collectively define an interior volume **60** of the diffuser assembly **50**. As discussed in detail below, the first wall **54** may include a wall flange **62** extending therefrom and defining a passage **64** through the first wall **54** and into the interior volume **60** of the housing **52**. The passage **64** is configured to receive a first flange **66** of a first adjustment panel **68** (e.g., a first adjustable panel) of the diffuser assembly **50** and to receive a second flange **70** of a second adjustment panel **72** (e.g., a second adjustable panel) of the diffuser assembly **50**. The first

flange **66** and the second flange **70** may define at least a portion of an inlet port **74** of the diffuser assembly **50**. That is, in some embodiments, an inner perimeter **78** of the inlet port **74** may be defined by the first flange **66**, the second flange **70**, and portions of the wall flange **62**, for example.

(29) A duct **80** of the ductwork **14** may include an end portion **82** (e.g., duct outlet) that is configured to extend into, extend about, couple to, or otherwise be fluidly connected to the inlet port **74**. In particular, the end portion **82** may be configured to couple to the first flange **66** of the first adjustment panel **68**, the second flange **70** of the second adjustment panel **72**, the wall flange **62**, or a combination thereof. As such, the duct **80** may direct a conditioned air flow (e.g., a cooled air flow, a heated air flow, a dehumidified air flow) that may be generated by the HVAC unit **12**, for example, through the inlet port **74** and into the interior volume **60** of the housing **52**. The housing **52** may discharge the conditioned air flow received from the duct **80** via an outlet port **83** of the housing **52**, which may be formed between the first wall **54** and the second wall **56**. In an installed configuration, the outlet port **83** may be exposed to or otherwise fluidly coupled to the space **34** of the room **30**. In this manner, the diffuser assembly **50** may facilitate distribution and/or dispersion of the conditioned air flow received from the duct **80** into the space **34** of the room **30**, for example.

(30) As discussed in detail below, the first adjustment panel **68** and the second adjustment panel **72** may be moveably coupled to the housing **52** and configured to translate along the first wall **54** (e.g., relative to the housing **52**) toward one another and/or away from one another. In particular, the first adjustment panel **68** and the second adjustment panel **72** may translate along an axis **90**, relative to the first wall **54**, where the axis **90** may extend generally parallel to a width **92** of the housing **52**. In this manner, cooperative adjustment of the first adjustment panel **68** and the second adjustment panel **72** along and relative to the first wall **54** enables adjustment of a size (e.g., a cross-sectional area, a cross-sectional shape) of the inlet port **74**. In particular, adjustment of the first and second adjustment panels **68**, **72** along the first wall **54** enables a size of the inlet port **74** to be increased or decreased to enable the inlet port **74** to adequately receive the end portion **82** of the duct **80**, based on the particular size of the duct **80** (e.g., a relatively large or wider duct **80**, a relative small or narrow duct **80**).

(31) Accordingly, in embodiments where the duct **80** has a size that is relatively large, the first adjustment panel **68** and the second adjustment panel **72** may be transitioned to a first configuration **96** (see FIG. **8**), in which the first flange **66** of the first adjustment panel **68** and the second flange **70** of the second adjustment panel **72** are positioned at opposing sides or edges of the passage **64** (e.g., with respect to the axis **90**), such that the inlet port **74** is relatively large and configured to receive and secure the relatively large duct **80**. Conversely, in embodiments where the duct **80** has a size that is relatively small, the first adjustment panel **68** and the second adjustment panel **72** may be transitioned to a second configuration **98** (see FIG. **11**), in which the first flange **66** of the first adjustment panel **68** and the second flange **70** of the second adjustment panel **72** are engaged with or abutting one another or positioned substantially adjacent one another (e.g., with respect to the axis **90**), such that the inlet port **74** is relatively small and configured to receive and secure the relatively small duct **80**. As such, it should be appreciated that adjustment of the size of the inlet port **74** may facilitate coupling the diffuser assembly **50** to various different sizes of the duct **80**. As discussed below, the first adjustment panel **68** and the second adjustment panel **72** may be transitioned to a plurality of configurations between the first configuration **96** and the second configuration **98**, such that the first flange **66** of the first adjustment panel **68**, the second flange **70** of the second adjustment panel **72**, and corresponding portions of the wall flange **62** may cooperate to adjust the size of the inlet port **74** to a plurality of different values.

(32) FIG. **4** is an exploded perspective view of an embodiment of the diffuser assembly **50**. In some embodiments, the first wall **54**, the second wall **56**, and the web **58** of the housing **52** may be separate components that are coupled to one another via fasteners, adhesives, and/or a metallurgical process, such as welding or brazing. In some embodiments, the wall flange **62** that may extend from and cross-wise to an outer surface **100** of the first wall **54**. The wall flange **62** may define an

inner perimeter **101** of the passage **64**. For example, in the illustrated embodiment, the wall flange **62** includes a set of profiled flange sections **102** (e.g., lateral flange sections, curved flange sections) and a set of flange mounting sections **104** (e.g., linear flange sections) that extend between the profiled flange sections **102**. As such, profiled flange sections **102** and the flange mounting sections **104** may collectively define the inner perimeter **101** of the passage **64**.

(33) In some embodiments, the flange mounting sections **104** may extend generally parallel to one another and along the axis **90**, for example. Either or both of the flange mounting sections **104** may include one or more first apertures **110** formed therein that, as discussed below, may be configured to facilitate securement of the first adjustment panel **68** and the second adjustment panel **72** to the first wall **54** at plurality of discrete positions. In some embodiments, each of the profiled flange sections **102** may be self-similar and have a semi-circular profile that extends between corresponding ends of the flange mounting sections **104**. To this end, the inner perimeter **101** of the passage **64** may include a generally oblong profile. However, although the passage **64** is illustrated as having a generally oblong cross-sectional profile in the illustrated embodiment of FIG. 4, it should be appreciated that, in other embodiments, the passage **64** may have any other suitable cross-sectional profile. For example, in some embodiments, the profiled flange sections **102** may be linear sections that extend between the flange mounting sections **104** to form a quadrilateral cross-sectional profile of the passage **64**, a trapezoidal cross-sectional profile of the passage **64**, or another suitable cross-sectional profile of the passage **64**. In certain embodiments, the wall flange **62** may be a component that is separate from the first wall **54** and coupled to the first wall **54** (e.g., via a metallurgical process). In other embodiments, the wall flange **62** may be formed integrally with the first wall **54**. For example, the first wall **54** may be formed from sheet metal that is stamped to include the wall flange **62** formed integrally therewith.

(34) For conciseness, the first adjustment panel **68** and the second adjustment panel **72** may be collectively referred to herein as adjustment panels **120** of the diffuser assembly **50**. In some embodiments, the first adjustment panel **68** and the second adjustment panel **72** may be self-similar components that are interchangeable with one another. Moreover, the first flange **66** of the first adjustment panel **68** and the second flange **70** of the second adjustment panel **72** may be collectively referred to herein as adjustment panel flanges **122**. In the illustrated embodiment of FIG. 4, the adjustment panels **120** each include a base panel **124** (e.g., a base portion) and the corresponding adjustment panel flanges **122** (e.g., flange portions), which may extend cross-wise to and from the base panels **124**. In some embodiments, the adjustment panel flanges **122** may each include a profiled portion **126** (e.g., a curved portion) and a set of mounting portions **128** (e.g., linear portions) that extend from the profiled portion **126**. The mounting portions **128** may include one or more second apertures **129** formed therein that, as discussed below, may be configured to align with corresponding ones of the first apertures **110** to facilitate securement of the first adjustment panel **68** and the second adjustment panel **72** to the first wall **54** at plurality of discrete positions along the first wall **54**.

(35) In some embodiments, the profiled portions **126** of the adjustment panel flanges **122** may include profiles that match (e.g., are geometrically similar too) or correspond with the profiled flange sections **102** of the wall flange **62**. In an installed configuration **130** (see FIG. 5) of the adjustment panels **120** on the first wall **54**, first sliding surfaces **132** of the base panels **124** may be configured to engage with (e.g., physically contact) a second sliding surface **134** of the first wall **54**, and the adjustment panel flanges **122** may extend through the passage **64**. Further, in the installed configuration **130** of the adjustment panels **120** on the first wall **54**, the mounting portions **128** of the adjustment panel flanges **122** may extend generally parallel to the flange mounting sections **104** of the wall flange **62**.

(36) The diffuser assembly **50** includes a plurality of upper guide tabs **140** and a plurality of lower guide tabs **142** that are configured to couple to the first wall **54** and, as discussed below, guide translation of the adjustment panels **120** along the axis **90**, relative to the first wall **54**. For example,

to better illustrate the engagement between the upper guide tabs **140** and the first wall **54**, FIG. 5 is a cross-sectional perspective view of an embodiment of a portion of the diffuser assembly **50**, illustrating the upper guide tabs **140** in an installed configuration **144** on the first wall **54**. For clarity, in the illustrated embodiment of FIG. 5, the adjustment panels **120** are in the installed configuration **130** on the first wall **54**. In some embodiments, the upper guide tabs **140** may each include a first mounting flange **150**, a first guide flange **152**, and a first web **154** (e.g., transition section, offsetting portion) that extends between the first mounting flange **150** and the first guide flange **152**. The first mounting flanges **150** may be coupled to the second sliding surface **134** of the first wall **54** in the installed configuration **144** of the upper guide tabs **140**. The first mounting flanges **150** may be coupled to the first wall **54** via fasteners, adhesives, or a metallurgical process, such as welding or brazing. In the installed configuration **144** on the first wall **54**, the upper guide tabs **140** may form a set of first channels **156** that extend between the second sliding surface **134** of the first wall **54** and corresponding surfaces the first guide flanges **152**. Each of the first channels **156** is configured to receive a respective one of the base panels **124** of the adjustment panels **120**. In this way, the upper guide tabs **140** may enable translation of the adjustment panels **120** along the first channels **156** and the axis **90**, relative to the first wall **54**, while substantially blocking movement of the adjustment panels **120** along an additional axis **158** that may extend parallel to a depth **160** of the housing **52**, for example. That is, the upper guide tabs **140** may guide translation of the first sliding surfaces **132** (see FIG. 4) of the adjustment panels **120** along the second sliding surface **134** of the first wall **54** (e.g., along the axis **90**).

(37) To better illustrate the engagement between the lower guide tabs **142** and the first wall **54**, FIG. 6 is a cross-sectional perspective view of an embodiment of a portion of the diffuser assembly **50**, illustrating the lower guide tabs **142** in an installed configuration **162** on the first wall **54**. For clarity, in the illustrated embodiment of FIG. 6, the adjustment panels **120** are in the installed configuration **130** on the first wall **54**. In some embodiments, the lower guide tabs **142** may each include a second mounting flange **164**, a second guide flange **166**, and a second web **168** (e.g., transition section, offsetting portion) that extends between the second mounting flange **164** and the second guide flange **166**. The second mounting flanges **164** may be coupled to the second sliding surface **134** of the first wall **54** in the installed configuration **162** of the lower guide tabs **142**. The second mounting flanges **164** may be coupled to the first wall **54** via fasteners, adhesives, or a metallurgical process, such as welding or brazing. In the installed configuration **162** on the first wall **54**, the lower guide tabs **142** may form a set of second channels **170** that extend between the second sliding surface **134** of the first wall **54** and corresponding surfaces the second guide flanges **166**. Each of the second channels **170** is configured to receive a respective one of the base panels **124** of the adjustment panels **120**. Thus, the lower guide tabs **142** may enable translation of the adjustment panels **120** along the second channels **170** and the axis **90**, relative to the first wall **54**, while substantially blocking movement of the adjustment panels **120** along the additional axis **158**, for example. That is, the lower guide tabs **142** may facilitate translation of the first sliding surfaces **132** (see FIG. 4) of the adjustment panels **120** along the second sliding surface **134** of the first wall **54** (e.g., along the axis **90**).

(38) FIG. 7 is an elevation view of an embodiment of a portion of the diffuser assembly **50**, illustrating the adjustment panels **120** in the first configuration **96** (e.g., a fully open or expanded configuration). For clarity, the second wall **56** is not shown with the housing **52** in the illustrated embodiment of FIG. 7 to better illustrate the adjustment panels **120**. In the first configuration **96**, the adjustment panels **120** may be positioned such that the adjustment panel flanges **122** are engaged with (e.g., physically contact, abut) the wall flange **62**. That is, the adjustment panel flanges **122** may engage with the profiled flange sections **102** of the wall flange **62**. As generally discussed above, the inner perimeter **78** of the inlet port **74** may be defined by inner surfaces of the adjustment panel flanges **122** and inner surfaces of at least a portion of the flange mounting sections **104**. As such, in the first configuration **96** (e.g., an expanded configuration) of the

adjustment panels **120**, the inner perimeter **78** of the inlet port **74** may be relatively large and substantially equal to the inner perimeter **101** of the wall flange **62**, for example. In certain embodiments, one or more fasteners **180** may extend through corresponding ones of the first apertures **110** in the wall flange **62** and the second apertures **129** in the adjustment panel flanges **122** to couple the adjustment panels **120** to the first wall **54** at the first configuration **96**. In other embodiments, the adjustment panels **120** may be coupled to the first wall **54** at the first configuration **96** via any other suitable technique. In any case, while the adjustment panels **120** are in the first configuration **96**, the inlet port **74** may include a relatively large size and, thus, be configured to receive and secure a relatively large duct **80**. That is, in the first configuration **96**, a cross-sectional area of the inlet port **74** may be substantially equal to a cross-sectional area of the passage **64**. To further illustrate the engagement between the fasteners **180** and the apertures **110**, **129** in the first configuration **96** of the adjustment panels **120**, FIG. **8** is a perspective view of an embodiment of the diffuser assembly **50**, in which the adjustment panels **120** are coupled to the first wall **54** (e.g., via the fasteners **180**) in the first configuration **96**.

(39) FIG. **9** is an elevation view of an embodiment of a portion of the diffuser assembly **50**, illustrating the adjustment panels **120** in an intermediate configuration **188**. For clarity, the second wall **56** is not shown with the housing **52** in the illustrated embodiment of FIG. **9** to better illustrate the adjustment panels **120**. To transition the adjustment panels **120** from the first configuration **96** to the intermediate configuration **188**, an operator (e.g., a service technician installing the diffuser assembly **50** at a jobsite) may slide the first adjustment panel **68** in a first direction **190** along the axis **90** and slide the second adjustment panel **72** in a second direction **192** along the axis **90**, opposite the first direction **190**. In particular, the operator may slide the adjustment panels **120** along the axis **90** until selected apertures **110**, **129** are appropriately aligned to enable insertion of the fasteners **180** and coupling of the adjustment panels **120** to the first wall **54** at the intermediate configuration **188**. As such, while the adjustment panels **120** are in the intermediate configuration **188**, the inlet port **74** may include a moderate size and, thus, be configured to receive and secure a moderately-sized duct **80**. That is, in the intermediate configuration **188**, a cross-sectional area of the inlet port **74** may be less than a cross-sectional area of the passage **64**. Indeed, as shown in the illustrated embodiment, in the intermediate configuration **188** of the adjustment panels **120**, the base panels **124** may occlude at least a portion of the passage **64** to block undesired air flow between the adjustment panels **120** and the first wall **54**.

(40) FIG. **10** is an elevation view of an embodiment of a portion of the diffuser assembly **50**, illustrating the adjustment panels **120** in the second configuration **98** (e.g., a contracted configuration). For clarity, the second wall **56** is not shown with the housing **52** in the illustrated embodiment of FIG. **10** to better illustrate the adjustment panels **120**. To transition the adjustment panels **120** from the intermediate configuration **188** to the second configuration **98**, the operator may slide the first adjustment panel **68** in the first direction **190** along the axis **90** and slide the second adjustment panel **72** in the second direction **192** along the axis **90**, opposite the first direction **190**. In particular, the operator may slide the adjustment panels **120** along the axis **90** until selected apertures **110**, **129** are appropriately aligned to enable insertion of the fasteners **180** and coupling of the adjustment panels **120** to the first wall **54** at the second configuration **98**. In some embodiments, the first adjustment panel **68** may engage (e.g., contact, abut) the second adjustment panel **72** in the second configuration **98** of the adjustment panels **120**. In any case, while the adjustment panels **120** are in the second configuration **98**, the inlet port **74** may include a relatively small size and, thus, be configured to receive and secure a relatively small duct **80**. That is, in the second configuration **98**, a cross-sectional area of the inlet port **74** may be less than a cross-sectional area of the passage **64**. Indeed, as shown in the illustrated embodiment, the base panels **124** may occlude at least a portion of the passage **64** in the second configuration **98** of the adjustment panels **120** to block undesired air flow between the adjustment panels **120** and the first wall **54**. To further illustrate the engagement between the fasteners **180** and the apertures **110**, **129**

in the second configuration **98** of the adjustment panels **120**, FIG. **11** is a perspective view of an embodiment of the diffuser assembly **50**, in which the adjustment panels **120** are coupled to the first wall **54** (e.g., via the fasteners **180**) in second configuration **98**.

(41) FIG. **12** is a perspective view of an embodiment of the first adjustment panel **68**, illustrating a set of mounting slots **200** of the first adjustment panel **68**. As discussed below, the mounting slots **200** may be configured to receive fasteners to facilitate coupling of the first adjustment panel **68** to the housing **52** at a plurality of user-selectable positions. The mounting slots **200** may extend through the base panel **124** and along a width **202** of the first adjustment panel **68**. In the illustrated embodiment of FIG. **12**, the first adjustment panel **68** does not include the mounting portions **128**. Indeed, as discussed below, the mounting slots **200** may be used to facilitate coupling (e.g., fixed coupling) of the first adjustment panel **68** to the housing **52** in lieu of the mounting portions **128**. As such, in the illustrated embodiment, distal ends of the first flange **66** may terminate at an edge **206** of the base panel **124**. However, it should be appreciated that, in other embodiments, the first adjustment panel **68** may include both the mounting slots **200** and the mounting portions **128**. Moreover, it should be understood that the second adjustment panel **72** may include some of or all of the features of the first adjustment panel **68** discussed herein.

(42) FIG. **13** is a perspective view of an embodiment of the diffuser assembly **50**, in which the first adjustment panel **68** and the second adjustment panel **72** each include corresponding mounting slots **200**. The diffuser assembly **50** may include a plurality of apertures **212** formed in the first wall **54** of the housing **52**. The apertures **212** may each be configured to align with a corresponding one of the mounting slots **200** formed in the first and second adjustment panels **68**, **72**. Fasteners **214** may extend through the apertures **212** and through a corresponding one of the mounting slots **200**. Retention features (e.g., nuts) may be coupled to ends of the fasteners **214** to wedge or capture the base panels **124** between the retention features and the second sliding surface **134** of the first wall **54**. For example, in some embodiments, the retention features may be adjusted such that the engagement between the retention features and the fasteners **214** applies a slight compressive force between the second sliding surface **134** of the first wall **54** and the base panels **124**, while still enabling translation of the first and second adjustment panels **68**, **72** along the first wall **54** in the first or second directions **190**, **192** (e.g., based on a force applied to the first and/or second adjustment panels **68**, **72** by the operator). Moreover, in this manner, engagement between the fasteners **214** and the mounting slots **200** may enable translation of the first adjustment panel **68** and the second adjustment panel **72** along the axis **90**, relative to the housing **52**, while substantially blocking movement of the first and second adjustment panels **68**, **72** along the additional axis **158**, relative to the housing **52**. As such, in some embodiments, the upper guide tabs **140** and the lower guide tabs **142** may be omitted from the diffuser assembly **50**.

(43) In any case, during, for example, installation of the diffuser assembly **50**, the operator may slide (e.g., independently slide) the first and second adjustment panels **68**, **72** along the housing **52** until the inlet port **74** reaches a size (e.g., cross-sectional area, cross-sectional shape) that is suitable to receive and secure the duct **80**. Once the first and second adjustment panels **68**, **72** are positioned in a manner that enables the inlet port **74** to adequately receive and couple to the duct **80**, the operator may tighten the retention features on the fasteners **214** to fixedly couple the first and second adjustment panels **68**, **72** to the housing **52** and block further or undesired movement of the first and second adjustment panels **68**, **72** relative to the housing **52**.

(44) FIG. **14** is a perspective view of another embodiment of the adjustment panels **120**, in which the adjustment panels **120** each include a single mounting portion **128**, referred to herein as an extended mounting portion **220**. In some embodiments, each of the extended mounting portions **220** may include a length **222** that is substantially equal to a length **224** (see FIG. **4**) of each of the flange mounting sections **104**. As such, when the adjustment panels **120** are installed on the first wall **54** and disposed in the first configuration **96**, the extended mounting portions **220** may extend across the lengths **224** of the flange mounting sections **104** and may be coupled to the flange

mounting sections **104** in accordance with the aforementioned techniques. Upon transitioning the adjustment panels **120** from the first configuration **96** toward the second configuration **98**, the extended mounting portions **220** may engage with corresponding inner surfaces **226** of the adjustment panel flanges **122** and translate along the inner surfaces **226**. In certain embodiments, upon transitioning the adjustment panels **120** to an appropriate or desired position on the first wall **54**, the operator may trim (e.g., cut shear) any portion of the extended mounting portions **220** that may overlap with the inner surfaces **226** from the first and/or second adjustment panels **68**, **72** (e.g., using tin shears). It should be appreciated that, in certain embodiments, the first adjustment panel **68**, the second adjustment panel **72**, or both, may include the mounting slots **200** in addition to the extended mounting portions **220**. As such, the first or second adjustment panels **68**, **72** may be coupled to the housing **52** in accordance with the techniques discussed above.

(45) As set forth above, embodiments of the present disclosure may provide one or more technical effects useful for coupling (e.g., fluidly coupling, mechanically coupling) an individual diffuser assembly to multiple different types of ducts to facilitate installation of the diffuser assembly at a jobsite, as well as reduce costs and/or complexity that may be associated with manufacturing the diffuser assembly. The technical effects and technical problems in the specification are examples and are not limiting. It should be noted that the embodiments described in the specification may have other technical effects and can solve other technical problems.

(46) While only certain features and embodiments have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, such as temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth, without materially departing from the novel teachings and advantages of the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure.

(47) Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual implementation may not have been described, such as those unrelated to the presently contemplated best mode, or those unrelated to enablement. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure, without undue experimentation.

(48) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

Claims

1. A diffuser assembly for a heating, ventilation, and air conditioning (HVAC) system, the diffuser assembly comprising: a housing comprising a wall and a passage formed in the wall; a first adjustment panel moveably coupled to the wall and comprising a first flange; and a second adjustment panel moveably coupled to the wall and comprising a second flange, wherein the first flange and the second flange define at least a portion of an inlet port of the diffuser assembly that

extends through the passage, wherein the inlet port is configured to fluidly couple to a duct and direct an air flow from the duct into an interior volume of the housing, and wherein the first adjustment panel and the second adjustment panel are configured to translate along the wall to adjust a size of the inlet port, wherein the wall comprises a mounting section extending from a surface of the wall and forming a portion of a perimeter of the passage, wherein the first flange comprises a mounting portion, and wherein the mounting portion of the first flange is configured to couple to the mounting section of the wall to secure the first adjustment panel to the wall.

2. The diffuser assembly of claim 1, wherein the first adjustment panel comprises a first base portion having a first sliding surface, the second adjustment panel comprises a second base portion having a second sliding surface, and the wall comprises a third sliding surface, wherein the first sliding surface and the second sliding surface are configured to engage with and translate along the third sliding surface.

3. The diffuser assembly of claim 2, wherein the first base portion and the second base portion are disposed within the interior volume of the housing, the first flange extends from the first base portion into the passage, and the second flange extends from the second base portion into the passage.

4. The diffuser assembly of claim 1, comprising a guide tab coupled to a first surface of the wall to form a guide channel between the first surface of the wall and a second surface of the guide tab, wherein the first adjustment panel comprises a base portion disposed within the guide channel, and wherein the guide tab is configured to guide translation of the base portion along the wall.

5. The diffuser assembly of claim 1, wherein the mounting portion comprises a first aperture formed therein, and the mounting section comprises a plurality of second apertures formed therein, wherein the first aperture and a corresponding aperture of the plurality of second apertures are configured to receive a fastener to secure the first adjustment panel to the wall at one of a plurality of discrete positions.

6. The diffuser assembly of claim 1, wherein the wall comprises a third flange extending from the surface and about the perimeter of the passage, wherein the mounting section forms a portion of the third flange.

7. The diffuser assembly of claim 1, wherein a first length of the mounting portion is substantially equal to a second length of the mounting section.

8. The diffuser assembly of claim 1, wherein the first adjustment panel comprises a mounting slot formed therein, the wall comprises an aperture formed therein, and wherein the aperture and the mounting slot are configured to receive a fastener to moveably couple the first adjustment panel to the wall.

9. The diffuser assembly of claim 8, wherein engagement between the fastener, the mounting slot, and the aperture enables movement of the first adjustment panel along a first axis extending along a width of the housing and substantially blocks movement of the first adjustment panel along a second axis extending cross-wise to the first axis.

10. The diffuser assembly of claim 1, wherein the first adjustment panel and the second adjustment panel are self-similar.

11. A diffuser assembly for a heating, ventilation, and air conditioning (HVAC) system, the diffuser assembly comprising: a wall having a passage extending therethrough; a first adjustment panel moveably coupled to the wall and comprising a first flange extending into the passage; and a second adjustment panel moveably coupled to the wall and comprising a second flange extending into the passage, wherein the first flange and the second flange define at least a portion of an inlet port extending through the passage and configured to fluidly couple to a duct, wherein the first adjustment panel and the second adjustment panel are configured to translate along the wall between a first configuration to define a first size of the inlet port and a second configuration to define a second size of the inlet port, wherein the wall comprises a third flange that defines a perimeter of the passage, wherein, in the first configuration, the first flange is configured to engage

a first section of the perimeter and the second flange is configured to engage a second section of the perimeter such that a first cross-sectional area of the inlet port is substantially equal to a second cross-sectional area of the passage.

12. The diffuser assembly of claim 11, wherein, in the second configuration, the first flange engages the second flange such that the first flange and the second flange are offset from the perimeter of the passage and define a third cross-sectional area of the inlet port that is less than the second cross-sectional area of the passage.

13. The diffuser assembly of claim 12, wherein the first adjustment panel and the second adjustment panel comprise base portions configured to engage with and translate along the wall, wherein, in the second configuration of the first adjustment panel and the second adjustment panel, the base portions are configured to occlude a portion of the passage.

14. The diffuser assembly of claim 11, wherein the inlet port is configured to physically couple to an end portion of the duct of the HVAC system.

15. A diffuser assembly for a heating, ventilation, and air conditioning (HVAC) system, the diffuser assembly comprising: a housing comprising a wall and a passage formed in the wall; a first adjustment panel comprising a first base portion moveably coupled to the wall, and comprising a first flange extending from the first base portion into the passage; and a second adjustment panel comprising a second base portion moveably coupled to the wall, and comprising a second flange extending from the second base portion into the passage, wherein the first flange and the second flange define at least a portion of an inlet port extending through the passage and configured to receive an air flow from a duct, and wherein the first base portion and the second base portion are configured to translate along the wall to expose or occlude a portion of the passage to adjust a size of the inlet port in the passage, wherein the wall comprises a mounting section extending from a surface of the wall and forming a portion of a perimeter of the passage, wherein the first flange comprises a mounting portion, and wherein the mounting portion of the first flange is configured to couple to the mounting section of the wall to secure the first adjustment panel to the wall.

16. The diffuser assembly of claim 15, wherein the housing comprises an additional wall and a web extending between the wall and the additional wall to define an interior volume of the housing, wherein the interior volume extends between a first inner surface of the wall and a second inner surface of the additional wall, and wherein the first base portion and the second base portion are disposed within the interior volume.

17. The diffuser assembly of claim 16, comprising a guide tab coupled to the first inner surface of the wall to form a channel between the first inner surface and a surface of the guide tab, wherein the first base portion is disposed within the channel, and wherein the guide tab is configured to guide translation of the first base portion along the first inner surface.

18. The diffuser assembly of claim 15, wherein the first base portion comprises a mounting slot formed therein and the wall comprises an aperture formed therein, wherein the aperture and the mounting slot are configured to receive a fastener to moveably couple the first base portion to the wall.

19. The diffuser assembly of claim 15, wherein a first length of the mounting portion is substantially equal to a second length of the mounting section.
